

PUERTO PLATA LUPERON, Dominican Republic

WMO: **784570**Lat: **19.758N**Lon: **70.570W**Elev: **5**StdP: **101.27**Time Zone: **-4.00 (W04)**Period: **00-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.2	19.9	17.7	12.7	21.5	18.3	13.2	21.5	8.5	27.6	8.2	27.3	0.5	180	0.313

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	33.9	27.0	33.1	26.8	32.8	26.8	28.1	32.0	27.7	31.7	27.4	31.4	6.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	22.8	31.2	26.7	22.3	30.9	26.2	21.6	30.4	90.0	32.1	88.3	32.0	86.9	31.8	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	8.0	DB	17.7	36.3	0.9	1.3	17.0	37.2	16.5	38.0	16.0	38.8	15.4	39.7	(4)
(5)			WB	16.5	29.2	1.2	1.2	15.6	30.1	15.0	30.8	14.3	31.5	13.5	32.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.5	24.3	24.5	24.8	25.8	26.7	28.1	28.3	28.4	28.2	27.7	26.2	25.2
	DBStd	1.84	1.05	1.07	1.33	1.09	1.07	0.97	0.90	0.93	0.85	0.99	1.09	1.05
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6032	444	407	458	473	519	542	568	570	546	548	487	471
	CDD18.3	2991	186	174	199	223	261	292	309	312	296	289	237	212
	CDH23.3	27346	1142	1134	1419	1784	2349	3149	3453	3504	3217	2886	1894	1415
CDH26.7	9703	246	255	359	519	787	1273	1403	1448	1346	1127	586	354	
Wind	WSAvg	3.2	2.6	3.1	3.3	3.5	3.5	3.7	4.0	3.8	3.1	2.7	2.5	2.5
Precipitation	PrecAvg	1773	219	147	150	154	123	48	58	78	70	134	294	295
	PrecMax	2352	567	316	479	367	457	121	184	214	167	374	554	814
	PrecMin	1077	49	2	8	6	14	5	11	20	16	7	57	44
	PrecStd	352	126	91	104	102	118	33	48	52	40	85	133	198
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	31.0	31.9	32.1	33.1	34.8	34.2	34.2	34.8	34.1	32.2	31.1
		MCWB	24.4	24.9	24.7	25.4	25.9	27.0	26.8	27.1	27.0	27.0	26.3	25.4
	2%	DB	29.8	29.9	30.2	31.0	32.0	33.2	33.2	33.2	33.2	33.0	31.2	30.1
		MCWB	24.6	24.7	24.9	25.6	26.1	26.8	26.7	27.0	27.1	26.9	26.4	25.2
	5%	DB	28.9	29.0	29.8	30.1	31.1	32.2	32.7	32.9	32.6	32.1	30.7	29.2
		MCWB	24.5	24.4	24.8	25.3	26.0	26.7	26.7	26.9	27.0	26.8	26.3	25.0
	10%	DB	28.1	28.2	28.8	29.2	30.2	31.9	32.0	32.1	32.0	31.2	29.9	28.8
		MCWB	24.1	23.9	24.3	24.9	25.8	26.6	26.5	26.8	26.8	26.6	25.8	24.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	26.2	26.3	27.0	27.6	28.6	28.2	28.4	28.5	28.2	27.5	26.6
		MCDB	29.0	29.5	29.7	30.3	30.7	32.2	32.1	32.4	32.5	32.0	30.3	29.2
	2%	WB	25.3	25.4	25.6	26.2	26.9	27.8	27.7	27.9	27.9	27.6	26.8	26.0
		MCDB	28.5	28.9	29.2	29.7	30.5	31.7	31.8	32.0	32.0	31.6	30.2	29.0
	5%	WB	24.9	24.8	25.0	25.6	26.5	27.2	27.2	27.5	27.5	27.2	26.5	25.5
		MCDB	28.1	28.3	28.8	29.3	30.1	31.3	31.4	31.7	31.6	31.1	29.8	28.6
	10%	WB	24.4	24.3	24.5	25.2	26.0	26.8	26.8	27.1	27.1	26.7	26.0	25.0
		MCDB	27.5	27.6	28.1	28.8	29.7	30.9	30.9	31.2	31.2	30.6	29.4	28.2
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.8	7.8	7.5	7.6	8.1	7.9	7.9	8.4	7.9	7.3	7.4
		MCDBR	9.0	8.9	9.1	8.5	8.5	8.8	8.7	8.7	9.2	8.9	8.4	8.3
		MCWBR	4.9	4.7	4.6	4.3	4.2	4.2	3.8	3.8	4.3	4.3	4.4	4.4
	5% WB	MCDBR	8.3	8.5	8.5	8.0	8.0	8.4	8.0	8.1	8.6	8.3	7.9	7.8
		MCWBR	5.0	4.9	4.7	4.4	4.4	4.5	4.1	4.1	4.5	4.4	4.5	4.5
Clear-Sky Solar Irradiance	taub	0.371	0.386	0.399	0.428	0.458	0.524	0.510	0.484	0.436	0.397	0.390	0.392	
	taud	2.552	2.494	2.454	2.397	2.333	2.099	2.141	2.240	2.412	2.528	2.538	2.503	
	Ebn at Noon	909	912	910	880	845	783	795	821	862	890	882	873	
	Edh at Noon	96	107	115	123	129	162	156	142	118	102	97	98	
All-Sky Solar Radiation	RadAvg	4.26	5.04	5.67	5.99	5.90	6.22	6.27	6.08	5.75	5.02	4.22	4.04	
	RadStd	0.30	0.29	0.39	0.43	0.39	0.32	0.32	0.26	0.36	0.37	0.35	0.23	

Historical Trends